



NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

Department of Mathematics

COMPUTATIONAL PROBLEM SOLVING AND DATA STRUCTURES

Assignment

Course: Computational Problem Solving and Data Structures**Instructor:** Dr. Sivalingam S M

1. Write a Python program that reads the number of electricity units consumed and calculates the electricity bill according to the following tariff:

- First 100 units: Rs. 2 per unit,
- Next 100 units: Rs. 3 per unit,
- Next 100 units: Rs. 5 per unit,
- Units above 300: Rs. 7 per unit.

If the calculated bill exceeds Rs. 1500, an additional surcharge of 5% of the bill must be added. Display the final amount payable.

2. Write a Python program to read the marks obtained by a student in five subjects. The program must:

- (a) calculate the total marks,
- (b) calculate the percentage,
- (c) determine whether the student has passed or failed,
- (d) assign an appropriate grade.

A student is considered to have failed if the mark in any one subject is below 40. For a student who passes, use the following grading scheme:

Percentage	Grade
90 and above	A
80–89	B
70–79	C
60–69	D
Below 60	E

3. Write a Python program to read three positive numbers representing the lengths of the sides of a triangle. The program should determine:

- (a) whether the three sides can form a valid triangle,
- (b) if valid, whether the triangle is equilateral, isosceles, or scalene,
- (c) whether the triangle is right-angled.

4. An ATM contains currency notes of denominations Rs. 500, Rs. 200 and Rs. 100. Write a Python program that reads the amount to be withdrawn. Determine the minimum number of notes required to dispense the requested amount. For example, for an amount of Rs. 1800, the program may display:

```
500 rupee notes = 3
200 rupee notes = 1
100 rupee notes = 1
```

5. A parking facility charges vehicles according to the following rule:

- First 2 hours: Rs. 30 per hour,
- Next 3 hours: Rs. 20 per hour,
- Every hour beyond 5 hours: Rs. 10 per hour.

If the vehicle remains parked for more than 10 hours, a discount of Rs. 50 is given on the total bill. Write a Python program that reads the number of hours for which the vehicle was parked and prints the final parking charge.

6. Write a Python program to read a positive integer and determine: a) the number of digits, b) the sum of its digits, c) the largest digit, and d) the smallest digit. The number must be processed using arithmetic operations and loops. For example, if the input is 58342, the program should obtain Number of digits = 5, Sum of digits = 22, Largest digit = 8, and Smallest digit = 2.

7. Write a Python program to read a positive integer. Using a loop:

- (a) reverse the number,
- (b) display the reversed number,
- (c) determine whether the original number is a palindrome.

8. A three-digit number is said to be special if the sum of the cubes of its digits is equal to the original number. For example,

$$153 = 1^3 + 5^3 + 3^3.$$

Write a Python program to print all such numbers between 100 and 999.

9. Write a Python program to read an integer $n > 1$ and determine whether it is prime or composite. If the number is composite, also print its smallest divisor greater than 1. For example, if the input is 91, the program should display:

```
Composite
Smallest divisor = 7
```

10. Write a Python program to read two integers a and b and print all prime numbers in the interval $[a, b]$. Your program should also print:

- (a) the total number of primes found,
- (b) the sum of all prime numbers in the interval.

11. A positive integer is called a perfect number if the sum of its proper divisors is equal to the number itself. For example,

$$6 = 1 + 2 + 3.$$

Write a Python program to:

- (a) determine whether a given positive integer is a perfect number,
- (b) print all perfect numbers between 1 and 10000.

12. Write a Python program to read a positive integer n . Repeatedly perform the following operations:

$$n = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even,} \\ 3n + 1, & \text{if } n \text{ is odd.} \end{cases}$$

Continue the process until $n = 1$. For example, 6, 3, 10, 5, 16, 8, 4, 2, 1. Your program must display:

- (a) the complete generated sequence,
- (b) the total number of steps,
- (c) the largest value encountered during the process.

13. Write a Python program to read an integer n and generate the following pattern. For $n = 5$:

```
1
12
123
1234
12345
```

14. Write a Python program to read an integer n and generate a symmetric number pattern. For $n = 5$, the required output is:

```
1
121
12321
1234321
123454321
```

15. Write a Python program that reads a positive integer n and prints an $n \times n$ multiplication table. For $n = 4$, the output should be of the form

```
1 2 3 4
2 4 6 8
3 6 9 12
4 8 12 16
```

16. Write a Python program to read a password. A password is considered valid if it satisfies all of the following conditions:

- it contains at least 8 characters,
- it contains at least one uppercase English letter,
- it contains at least one lowercase English letter,
- it contains at least one digit.

The program should print whether the password is valid. If the password is invalid, the program should also identify which of the above requirements are not satisfied.

17. Write a Python program to read a sentence and separately count the number of a) uppercase letters, b) lowercase letters, c) digits, d) spaces, and e) other characters. For example, consider the input:

Room 204 is Ready!

The program should inspect every character in the sentence and report the corresponding counts.

18. Write a Python program to read: a) a sentence, and b) a character. Determine:

- how many times the character occurs in the sentence,
- all positions at which the character occurs.

19. Write a Python program to read a word or phrase and determine whether it is a palindrome. While performing the check ignore spaces, and ignore differences between uppercase and lowercase letters. For example,

`Never odd or even`

should be identified as a palindrome.

20. Write a Python program to read a string and print each group of consecutive identical characters together with the number of times that character occurs consecutively. For example, if the input is

```
aaabbccccdaa
```

the output should be

```
a 3 b 2 c 4 d 1 a 2
```

21. Write a Python program to read a string containing only lowercase English letters. Determine the character that occurs most frequently. If more than one character has the same maximum frequency, print all such characters. Your program should not assume beforehand which character occurs most often.

22. The hourly temperature readings recorded by a sensor are stored in a list. For example,

```
temperature = [32, 34, 31, 38, 41, 43, 39, 35, 33, 44]
```

Write a Python program to determine:

- (a) the maximum temperature,
- (b) the minimum temperature,
- (c) the average temperature,
- (d) the number of readings greater than 40°C,
- (e) the positions at which readings greater than 40°C occur.

23. Consider the following list:

```
data = [4, 7, 4, 2, 7, 9, 2, 8, 4]
```

Write a Python program to create a new list containing each value exactly once, while preserving the order in which the values first appeared. The required output is

```
[4, 7, 2, 9, 8]
```

24. Write a Python program to determine the second largest distinct value in a list of integers. For example,

```
data = [8, 3, 12, 12, 5, 9]
```

should produce

```
Second largest = 9
```

The program must correctly handle repeated maximum values.

25. Write a Python program that takes a list of integers and creates three new lists containing:

- (a) positive numbers,
- (b) negative numbers,
- (c) zeros.

26. The daily sales of a shop are stored in a list. For example,

```
sales = [120, 135, 128, 150, 165, 160, 190]
```

Write a Python program to:

- (a) identify all days on which the sales were greater than the previous day,
- (b) print the increase for each such day,
- (c) determine the largest increase between any two consecutive days.

For example, part of the output should look like:

```
Day 2 increased by 15
Day 4 increased by 22
```

27. Consider the list

```
data = [10, 20, 30, 40, 50, 60]
```

Write a Python program to read an integer k and rotate the list to the right by k positions. For example, if $k = 2$, the resulting list should be

```
[50, 60, 10, 20, 30, 40]
```

Your program should also work correctly when k is larger than the length of the list.

28. The marks obtained by a group of students are stored in the following dictionary:

```
marks = {"Anil": 76, "Bhavna": 92, "Charan": 68, "Divya": 92, "Farhan": 81}
```

Write a Python program to:

- (a) determine the highest mark,
- (b) print the names of all students who obtained the highest mark,
- (c) calculate the average mark,
- (d) print the names of all students whose marks are above the average.

29. Consider the list

```
data = [2, 4, 2, 5, 4, 2, 7, 5, 4, 4]
```

Write a Python program to construct a dictionary in which:

- each key is a number appearing in the list,
- the corresponding value is the number of times that number occurs.

For the above data, the resulting dictionary should be of the form

```
{2: 3, 4: 4, 5: 2, 7: 1}
```

30. A supermarket stores product prices in a dictionary as follows:

```
prices = {"rice": 60, "milk": 30, "bread": 40, "eggs": 6, "apple": 15}
```

Write a Python program that repeatedly asks the customer to enter the product name, and the required quantity. The customer enters

done

as the product name when shopping is complete. The program must:

- (a) verify whether the requested product exists in the dictionary,
- (b) calculate the cost of every requested item,
- (c) calculate the total bill,
- (d) calculate the total number of items purchased,
- (e) apply a 5% discount if the total bill exceeds Rs. 500,
- (f) display the final amount payable.

If a product is not available, print an appropriate message and continue accepting further input. A possible interaction is:

```
Product: rice
Quantity: 2

Product: apple
Quantity: 5

Product: soap
Product not available

Product: done
```

Finally, display the complete bill amount and the final amount payable after any applicable discount.